

# GEOOptimize

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| Proposal for:                                                                                      | Proposal Date: | Proposal Number:             |
|----------------------------------------------------------------------------------------------------|----------------|------------------------------|
| Bellaire AZ, LLC<br>11411 E. Arabian Park Drive<br>Scottsdale, AZ<br>85259<br><br>ATTN: Jeff Smith | 2024-02-20     | QU-0140                      |
|                                                                                                    |                | Reference                    |
|                                                                                                    |                | US AZ Orrion Farms – Phase 2 |

## Understanding of Project

Bellaire AZ has requested a proposal for a review of the design of a hybrid ground source heat pump (GSHP) system project located in Scottsdale Arizona. The project is a 18,900 ft<sup>2</sup> residence at 11411 E Arabian Park Drive.

A hybrid GSHP system is currently being constructed. GEOOptimize will work with the project team to ensure the proposed system will work sustainably and efficiently over the life of the system. After reviewing the system GEOOptimize will provide recommendations for changes to the hybrid system design and system control strategy that will improve the short and long-term efficient performance of the system if appropriate

## Proposed Design Methodology

### Hybrid Geothermal System Modelling

To model and design the configuration of the hybrid GSHP system, TRNSYS V18 will be used. TRNSYS is a simulation program used to model and optimize the performance of hybrid energy systems. Energy loads for the 18,900 ft<sup>2</sup> residence will be imported into the TRNSYS model to simulate the building loads and their impact on the hybrid GSHP system. The energy loads developed in Ekotrope will be used for this analysis due to the uncertainty of the building occupancy and the high variability of building energy loads based on building occupancy. TRNSYS software is used to analyse the performance of the GHX, building and the fluid coolers. TRNSYS models the capacity of the fluid cooler(s) based on Scottsdale, AZ hourly weather data to estimate the amount of heat that can be dissipated from the GHX during cooler winter and night-time temperatures. This maximizes the capacity of the fluid cooler(s). Using the fluid cooler(s) during cooler weather pre-cools the GHX to maximize the cooling capacity of the GHX throughout the year. Dissipating heat from the GHX through the year avoids long-term temperature increase in the vertical boreholes. Using the fluid coolers in cooler temperature also reduces water use for evaporation. A series of simulations of the model are developed to optimize system performance including:

- The size and configuration of GHX
- Heat dissipation from fluid coolers or cooling towers
- System control strategy

The TRNSYS modelling will determine the capacity of the GHX based on maximizing the capacity of the fluid cooler, optimizing the control strategies and provide monitoring solutions for the hybrid GSHP system.

## Sequence of Operations

The thermal dynamics of the fluid within a GHX system exhibit significant variations throughout the year. These fluctuations are primarily driven by the energy absorption or dispersion by the GHX, in conjunction with the thermal characteristics of the surrounding soil. The implementation of G-function algorithms facilitates the ongoing prediction of future GHX fluid temperatures, using real-time data on energy transfer to and from the GHX. Optimizing system performance, especially in preparation for the summer season, can be achieved by using a fluid cooler to 'precool' the GHX during winter months. Prediction of GHX fluid temperatures enables the management of the GHX temperatures, maximizing the long-term efficiency of the system, ensuring an efficient performance of the GHX throughout its operational lifespan.

In warmer locations such as Scottsdale, Arizona, where high temperatures are prevalent, the default use of a fluid cooler may not be the most effective strategy. Anticipating GHX temperatures empowers the strategic utilization of the fluid cooler during the winter, which maximizes the system coefficient of performance. Scottsdale typically experiences outdoor temperatures below 50°F for approximately five months each year (refer to Figure 1 for details). During the cooler season, the use of a dry fluid cooler can be calculated and automated. This proactive 'precharging' of the GHX for the upcoming summer season not only enhances system efficiency but also avoids the need for water in evaporative fluid coolers. By synchronizing the operation of the dry fluid cooler with colder external temperatures, energy consumption is optimized, and water usage is completely avoided, aligning with sustainable energy practices.

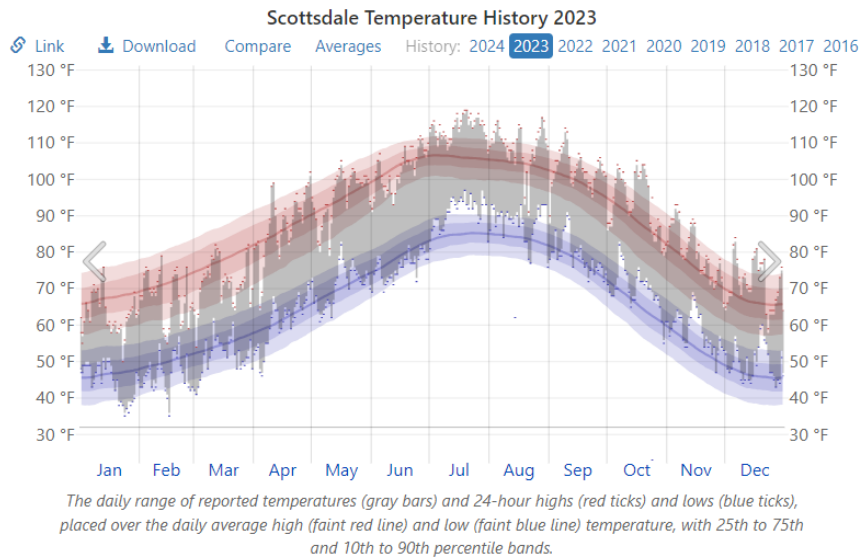


Figure 1: 2023 outdoor air temperature in Scottsdale Arizona.

## Scope of work

### Phase 2 – Recommendations

- Analysis of recommended system performance using TRNSYS to model the hybrid GSHP system.
- Recommend supply / return runout piping from GHX headers to manifold.
- Develop single line drawing of recommended mechanical piping system.
- Conceptual control strategy including long-term monitoring recommendations.

## Deliverables

### Phase 2 – Recommendations

- Single line drawing of recommended mechanical piping system.
- Findings and recommendations will be summarized in a PowerPoint presentation.
- Coordination via screen sharing sessions and possible site visits.

## Pricing

| Description | Line Total  |
|-------------|-------------|
| Phase 2     | \$26,100.00 |
| Total (USD) |             |

## Terms

- Additional work out of scope will be charged at \$250/hr.
- Invoices will be submitted monthly for work completed to date.
- Payment is expected within 30 days of invoice.
- Site visits will be charged at cost plus 5%

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